



Unit 3 · Outcome 1 · Maintaining Biodiversity

Sampling species diversity

KK06

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 3.1.2.6 Key science skill: generate & record data Lincoln index provided in exam 3D · drag to orbit A 0.25 m² quadrat dropped on the swamp floor at Kestrel Ridge Reserve

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



THE BIG IDEA

You measure a slice, then reason about the whole

A full census — counting every individual — is only possible for big, rare, obvious things (the entire wild population of Helmeted Honeyeaters is small enough to band one by one). For everything else, ecologists take a representative sample: measure a small, fairly chosen part of the habitat, then scale up.

◆ ANALOGY

Think of tasting a pot of soup. You don't drink the whole pot to judge it — you stir it well (mix it so the sample is representative), then taste one spoonful. The trick is making sure that spoonful really reflects the pot. Sampling a forest is the same: stir out the bias, then taste carefully.

How many individuals of a species are present (a raw count, or scaled to the whole area).

Individuals per unit area, e.g. 12 sedges / m². Count ÷ area sampled.

The % of the ground a species covers. Used for plants you can't count as individuals (grasses, mosses).

The % of quadrats a species turns up in. Present in 30 of 50 quadrats → 60% frequency.

The number of different species found — the headline biodiversity figure (feeds into Simpson's Index, KK07).

A sample whose composition mirrors the whole habitat — the goal of random placement and enough samples.

✓ TIP

Why it matters here Before you can maintain biodiversity at Kestrel Ridge, you need baseline numbers: how many honeyeaters, how dense the understorey sedge, which weeds are creeping in. Every management decision in KK11–KK13 rests on numbers that come from sampling.

TECHNIQUE 1

Quadrats — a frame of known area

A quadrat is a frame enclosing a known area (a classic one is 0.5 m × 0.5 m = 0.25 m²). You place it, then inside it you can record abundance, percentage cover, or which species are present. Because the area is fixed, you can convert a count into a density and scale it up to the whole habitat.

Below is a patch of swamp floor divided into a sampling grid. Drop quadrats and watch the estimate of total sedge plants settle down as your sample grows — this is sampling reliability happening in front of you.

◆ NOTE

Read the convergence One quadrat can be wildly off — you might land on a bare patch or a thicket. As you add quadrats, the mean density averages out the lucky and unlucky drops, and the estimate closes in on the true total. More samples → more reliable estimate. That's the whole argument for sample size in one animation.

The study design says "different-shaped quadrats" deliberately. Squares are common, but ecologists also use rectangles and circles. Shape changes how much edge you create per unit area — and edge is where counting errors live (next section). A long thin rectangle catches more of a gradient; a circle has the least edge for its area.

TECHNIQUE 2

Grids & the case for randomness

A grid divides the study area into equal numbered cells. You then choose which cells to sample. How you choose is the part examiners care about:

Use a random number generator to pick grid coordinates. Removes the human urge to pick "interesting" spots, so the sample is unbiased and represents the whole area.

Sample at fixed intervals (every 5th cell, or quadrats spaced along a line). Best when you want to capture change across a gradient rather than an overall average.

◆ NOTE

The bias trap "We placed quadrats where the sedge looked healthiest" is a biased method — it inflates the density and makes the reserve look better than it is. If a question asks how to improve validity, the magic words are random placement (via random number coordinates on a grid) and a larger sample size.

◆ ANALOGY

Throwing a quadrat over your shoulder "to be random" feels fair but isn't — you unconsciously avoid the prickly bushes and the mud. Random number coordinates on a grid are genuinely unbiased: the dice don't care where the mud is.

TECHNIQUE 3

Transects — sampling along a gradient

A transect is a line laid across a habitat. You use it when species composition changes across an environmental gradient — for example, from the wet creek edge at Kestrel Ridge, up the bank, to the dry ridge top. A random scatter of quadrats would blur that change; a transect captures it in order.

Record every species that touches the line at set points. Quick; good for a first picture of zonation.

Place quadrats at intervals along the line (a strip of known width). Gives abundance/cover data at each step of the gradient.

The belt transect below runs from the wet creek (left) to the dry ridge (right). Press play to walk the quadrats up the slope and watch how the community changes — moisture-loving sedge fades out, dry-tolerant tussock takes over.

✓ TIP

Decision rule Use a transect when the question mentions a gradient, zonation, distance from a feature, or "how does X change across...". Use random quadrats on a grid when it asks for an overall density or richness of a fairly uniform area.

THE DETAIL EVERYONE FORGETS

Edge effects — the boundary problem

Real organisms straddle the frame. If you count every plant touching the quadrat, you over-count (you're stealing from the neighbouring area too). If you count none of them, you under-count. Without a rule, two students sampling the same quadrat get different numbers — the data isn't repeatable.

✓ TIP

The convention Count an individual only if it touches two chosen sides (say the top and the left). Ignore anything touching the other two sides. Over many quadrats the overcounts and undercounts cancel, and everyone counts the same way — consistent and repeatable.

◆ NOTE

Shape & edge "Different-shaped quadrats" links straight to edge effects: a long thin rectangle has a high edge-to-area ratio, so more plants land on the boundary and edge-rule errors matter more. A circle has the lowest edge for its area. Mention this if a question asks why quadrat shape affects results.

TECHNIQUE 4 · FOR ANIMALS THAT MOVE

Mark-recapture & the Lincoln index

✦ RECAP / CHEAT SHEET

You can't put a quadrat around a Helmeted Honeyeater — it flies off. For mobile animals, ecologists estimate population size by capturing, marking, releasing, letting them mix, then recapturing:

- Capture & mark. Catch a first sample, mark them harmlessly (leg band), record the number M , release.
- Mix. Wait long enough for marked birds to disperse back through the population.
- Recapture. Catch a second sample of total size C , and count how many carry a mark, R .

◆ ANALOGY

Tip 100 raffle tickets into a crowd and ask everyone to keep one ($M = 100$ marked). Later you survey 200 people ($C = 200$) and 8 of them have a ticket ($R = 8$). The marked fraction in your survey ($8/200$) should match the marked fraction of the whole crowd ($100/N$). Rearranged: $N = (100 \times 200) / 8 = 2\,500$ people.

✦ RECAP / CHEAT SHEET

The Lincoln index is only valid if a set of assumptions hold. If a question asks why an estimate might be inaccurate, these are your answer bank:

Assumption	If it's broken...
Closed population — no births, deaths, immigration or emigration between captures	New unmarked individuals dilute the marked fraction → overestimate
Marks don't affect survival or behaviour	"Trap-shy" birds avoid recapture (R too low → overestimate); "trap-happy" birds return for the bait (R too high → underestimate)
Marks aren't lost between captures	Lost bands lower R → overestimate
Marked individuals mix randomly back in	If they don't disperse, recapture isn't representative
Enough time to mix, but not so long the population changes	Too short → clumped; too long → closed-population assumption fails

PUTTING IT TOGETHER

Which technique, and when?

If you're sampling...	Use...	Because
Plants / sessile organisms over a fairly uniform area	Random quadrats on a grid	They don't move; random placement gives an unbiased density & richness
How a community changes across a gradient (wet→dry, near→far)	Belt transect (quadrats along a line)	Captures zonation in order; random quadrats would blur it
A quick picture of zonation	Line transect	Fast record of what touches the line
Mobile animals (birds, possums, fish)	Mark–recapture (Lincoln index)	Can't be quadratted; estimate from the marked fraction

✓ TIP

One-line memory hook Rooted & random → quadrats. Gradient → transect. Moving → mark–recapture.

GOOD VS BAD · 4 MARKS

What a full-mark answer looks like

An ecologist uses mark–recapture to estimate the number of a small fish in a reserve wetland. They mark 60 fish, release them, and three days later catch 80 fish of which 15 are marked. Estimate the population and explain one reason the true number may be higher than your estimate. (4 marks)

• NOTE

Right ballpark by luck, but no formula shown, no working, and the "explanation" just restates the question without linking to an assumption.

• NOTE

Formula stated, values substituted, answer with units, and the explanation links a named assumption (closed population) to its effect on R and therefore on N.

◆ NOTE

The P-E-L move Point: $N = 320$ fish. Explain: the index assumes a closed population. Link: deaths/emigration lower R, so the marked fraction understates the truth and the real population is larger. Marks are earned for the chain, not the number alone.

DECODE THE QUESTION

Command words in this topic

◆ COMMAND WORDS

The verb tells you how much to write. Tap each one.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

ACTIVE RECALL

Sort the technique

✓ TIP

Nice All sorted. If any landed in red, re-read the decision rule: rooted & random → quadrats · gradient → transect · moving → mark-recapture.

EXAM PRACTICE · SELF-MARK · 18 MARKS

Practice questions

★ PRACTICE & SELF-MARK

Write your answer, reveal the marking scheme, then award yourself marks honestly — the dial bottom-right tracks your total. These are written in VCAA style for the Maintaining Biodiversity outcome.

WRITING TRAINER

Build a P·E·L answer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

Sampling questions reward a tight chain: make the Point, Explain the mechanism, then Link back to the question. Try one here.

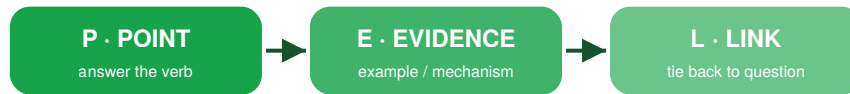
UNIT 3 · OUTCOME 1 · MAINTAINING BIODIVERSITY · KK06

Sampling species diversity

You can't count every honeyeater in the forest or every sedge on the swamp floor. So how do ecologists turn a handful of careful measurements into a believable number for the whole reserve? With grids, transects, quadrats and mark-recapture — and a healthy respect for the edges.

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

You do 60 times 80 which is a lot, then look at the 15 marked ones. There were about 400 fish. The number could be higher because some fish died or there were more fish than they thought. " Right ballpark by luck, but no formula shown, no working, and the "explanation" just restates the question without linking to an assumption.

✓ STRONG / MODEL ANSWER

$N = (M \times C) / R = (60 \times 80) / 15 = 320$ fish. The true number may be higher because the population is assumed to be closed, but if some marked fish died or emigrated between captures, fewer marks remain (R falls). A smaller R inflates the estimate downward relative to truth — wait, the marked fraction is too low, so the true population is larger than estimated." Formula stated, values substituted, answer with units, and the explanation links a named assumption (closed population) to its effect on R and therefore on N.